CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 4: Expansions

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Expand $(5a - 2b)^2$.
2. Evaluate $(105)^2$ and 96×104 using identities.
3. If $a + b = 9$ and $ab = 20$, find $a^2 + b^2$.
4. If $a - \frac{1}{a} = 3$, find $a^2 + \frac{1}{a^2}$.
5. If $x + \frac{1}{x} = 3$, find $x^3 + \frac{1}{x^3}$.
6. Expand $(x + 2y - 3z)^2$.
7. If $a + b + c = 12$ and $ab + bc + ca = 47$, find $a^2 + b^2 + c^2$.
8. Expand $(3a + 2b)^3$.
9. If $a - b = 3$ and $ab = 10$, find $a^3 - b^3$.
10. Evaluate $(15)^3 + (-9)^3 + (-6)^3$ without cubing.

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Answer Key

1. $25a^2 - 20ab + 4b^2$
2. 11025 and 9984
3. 41
4. 11
5. 18
6. $x^2 + 4y^2 + 9z^2 + 4xy - 12yz - 6zx$
7. 50
8. $27a^3 + 54a^2b + 36ab^2 + 8b^3$
9. $27 + 90 = 117$
10. $3 \times 15 \times (-9) \times (-6) = 2430$

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